

CSMFAB0000000000008 Aegis Home_ Carrington Storm Safe Cook-ware

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMFAB0000000000008 Aegis Home_ Carrington Storm Safe Cookware V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
YInMn Blue spectral coating	V1.0: YInMn + 2 mol% Ta2O5; SRI 115; NIR 85%	V2.0: CsPbBr3 QD co-deposit; SRI ~130; UV reflectance +31%	UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire	Solar Energy Mater. Sol. Cells 258 (2025) 112456
ZrB2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
YInMn Blue spectral coating	UV band protection gap; accelerated UV-induced substrate polymer degradation	MEDIUM
ZrB2-SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB0000000000008 Aegis Home_ Carrington Storm Safe Cook-ware V1.0 archived; V2.0 is the active specification as of June 2026.